18 Here are the first five terms of a quadratic sequence.

1 8 21 40 65

The nth term of this sequence can be written in the form $an^2 + bn$, where a and b are integers.

Work out the value of a and the value of b.

First differences: 7, 13, 19, 25

Second differences: 6, 6, 6

2a = 6, so a = 3

First term: a + b = 1

3 + b = 1, so b = −2

1 mark
Second difference ÷ 2.

1 mark
Substitute a = 3.

1 mark
a = 3, b = −2.

$a =$ .....

$b =$ .....

(Total for Question 18 is 3 marks)